



Vision Based Spraying

Caleb Stuber
Precision Planting



Solutions for Every Season





Why Do Farmers Spray?

- To control unwanted organisms that will cause crop loss
- Most commonly these would be weeds, fungi, and insects.

Why Do Farmers Spray?

- Weeds compete with the crop for Sunlight, Water, & Nutrients
- Weeds can reduce grain quality
- Weeds can act as a source or attractant for crop damaging insects and diseases
 - It is estimated that 18% of annual crop product is destroyed by insects
 - Sharma, S., Kooner, R., Arora, R. (2017) *Insect Pests and Crop Losses*
 - Fungal pathogens cause 10 – 20% crop losses each year worth \$100-\$200 billion annually.
 - U.S. Department of Agriculture, Agriculture Research Service (n.d.). *Food security: How do crop plants combat pathogens?*

Why Do Farmers Spray



Why Do Farmers Spray Weeds?

- Compounding Problem Through Weed Escapes
- Example: Waterhemp
 - Waterhemp will produce 28,000 – 160,000 viable seeds by a single female plant
 - 100,000 seeds/female plant = 50,000 Viable Seeds/Plant Average
 - 25% annual germination rate
 - ~5% of seeds will remain dormant and viable for up to 17 years

“Waterhemp - Weed of the Year.” *NDSU Weed Science*, www.ag.ndsu.edu/weeds/weed-of-the-year-files/waterhemp.

Loux, Mark. “Palmer Amaranth and Waterhemp Management: It’s All about the Seed.” *Ohioline*, Ohio State University Extension, 20 June 2019, ohioline.osu.edu/factsheet/anr-76.

Mohler, Charles L., et al. *Manage Weeds on Your Farm: A Guide to Ecological Strategies*. Sustainable Agriculture Research and Education Sustainable Agriculture Research and Education (SARE), 2021.

Seipel, Tim, and Lovreet Shergill. “Waterhemp (*Amaranthus Tuberculatus*): Biology, Identification, and Management.” *MSU Extension*, June 2022, apps.msuextension.org/montguide/guide.html?sku=MT202211AG.

Waterhemp Escape

- **Year 1: Weed Escape of One Waterhemp every 50'x50' (2,500sq ft)**
17 Waterhemp/A x 50,000 Viable Seeds/Plant = ***850,000 Seeds/A**
- **Year 2: 25% Germination and 99.98% Weed Control (1 in 5,000 Survive)**
850,000 Seeds/A x 25% Germination = 212,000 Weeds / 5,000 = **42 Waterhemp/A**
42 Waterhemp/A x 50,000 Seeds/Plant = 2.1 MILLION Seeds/A

Waterhemp Escape

- **Year 1: Weed Escape of One Waterhemp every 50'x50' (2,500sq ft)**
17 Waterhemp/A x 50,000 Viable Seeds/Plant = ***850,000 Seeds/A**
- **Year 2: 25% Germination and 99.98% Weed Control (1 in 5,000 Survive)**
42 Waterhemp/A **2.1 MILLION Seeds/A**
- **Year 3: 25% Germination and 99.99% Weed Control (1 in 10,000 Survive)**
2,100,000 Seeds/A x 25% Germination = 525,000 Weeds / 10,000 = **53 Waterhemp/A**
53 Waterhemp/A x 50,000 Seeds/Plant = 2.65 MILLION Seeds/A

Waterhemp Escape

- **Year 1: Weed Escape of One Waterhemp every 50'x50' (2,500sq ft)**
17 Waterhemp/A x 50,000 Viable Seeds/Plant = ***850,000 Seeds/A**
- **Year 2: 25% Germination and 99.98% Weed Control (1 in 5,000 Survive)**
42 Waterhemp/A **2.4 MILLION Seeds/A**
- **Year 3: 25% Germination and 99.99% Weed Control (1 in 10,000 Survive)**
53 Waterhemp/A **2.65 MILLION Seeds/A**

Waterhemp Escape

- **Year 1: Weed Escape of One Waterhemp every 50'x50' (2,500sq ft)**
17 Waterhemp/A x 50,000 Viable Seeds/Plant = ***850,000 Seeds/A**
- **Year 2: 25% Germination and 99.98% Weed Control (1 in 5,000 Survive)**
42 Waterhemp/A **2.4 MILLION Seeds/A**
- **Year 3: 25% Germination and 99.99% Weed Control (1 in 10,000 Survive)**
53 Waterhemp/A **2.65 MILLION Seeds/A**
- **Year 4: 25% Germination and 99.99% Weed Control (1 in 10,000 Survive)**
66 Waterhemp/A **3.3 MILLION Seeds/A**

Herbicide Global Consumption

- Global Cropland is approximately 3.7 billion acres
- Roughly 1.8 million metric tons of active ingredients

Cost/Acre To Raise Corn In Illinois

Category	\$/Acre	% of Total
Cash Rent (Land)	\$280	~23%
Fertilizers (Soil Fertility)	\$219	~18%
Seed	\$128	~10%
Pesticides	\$111	~9%
Machinery Depreciation	\$85	~7%
Machine Hire/Lease	\$95	~8%
Machine Repair	\$40	~3%
Fuel & Oil	\$36	~3%
Crop Insurance	\$24	~2%
Drying	\$24	~2%
Hired Labor	\$23	~2%
Interest (non-land)	\$20	~2%
Building & Storage, Utilities, Misc.	\$134	~11%
Total	\$1,221 per acre	

Source: farmdoc daily
(15): 19

Cost/Acre To Raise Corn In Illinois

Category	\$/Acre	% of Total
Cash Rent (Land)	\$280	~23%
Fertilizers (Soil Fertility)	\$219	~18%
Seed	\$128	~10%
Pesticides	\$111	~9%
Machinery Depreciation	\$85	~7%
Machine Hire/Lease	\$95	~8%
Machine Repair	\$40	~3%
Fuel & Oil	\$36	~3%
Crop Insurance	\$24	~2%
Drying	\$24	~2%
Hired Labor	\$23	~2%
Interest (non-land)	\$20	~2%
Building & Storage, Utilities, Misc.	\$134	~11%
Total	\$1,221 per acre	

Source: farmdoc daily
(15):19

Cost/Acre To Raise Corn In Illinois

Category	\$/Acre	% of Total
Cash Rent (Land)	\$280	~23%
Fertilizers (Soil Fertility)	\$219	~18%
Seed	\$128	~10%
Pesticides	\$111	~9%
Machinery Depreciation	\$85	~7%
Machine Hire/Lease	\$95	~8%
Machine Repair	\$40	~3%
Fuel & Oil	\$36	~3%
Crop Insurance	\$24	~2%
Drying	\$24	~2%
Hired Labor	\$23	~2%
Interest (non-land)	\$20	~2%
Building & Storage, Utilities, Misc.	\$134	~11%
Total	\$1,221 per acre	

Source: farmdoc daily
(15):19

That Which Does Not Kill You Makes You Stronger

Does cutting herbicide rates threaten the sustainability of weed management in cropping systems?

May 2011 · Journal of Theoretical Biology 283(1):14-27

283(1):14-27

DOI: [10.1016/j.jtbi.2011.05.010](https://doi.org/10.1016/j.jtbi.2011.05.010)

Source · [PubMed](#)

Authors:



Michael Renton
University of Western Australia



Arthur J Diggle
Department of Agriculture and



Sudheesh Manali
University of Western Australia



Stephen B Powles
University of Western Australia



ELSEVIER

Field Crops Research

Volume 292, 1 March 2023, 108819

Weed resistance to different herbicide modes of action is driven by agricultural intensification

Philip E. Hulme¹

Show more

Add to Mendeley Share Cite

<https://doi.org/10.1016/j.fcr.2023.108819>

Get right

Cutting herbicide rates spells trouble long-term

JUNE 18, 2018 | 4 MIN READ

Weeds Are Winning the War against Herbicide Resistance

Herbicides are under evolutionary threat. Can modern agriculture find a new way to fight back?

BY BROOKE BOREL



United Soybean Board Flickr (CC BY 2.0)



Fighting Back against Prolific Weed Resistance

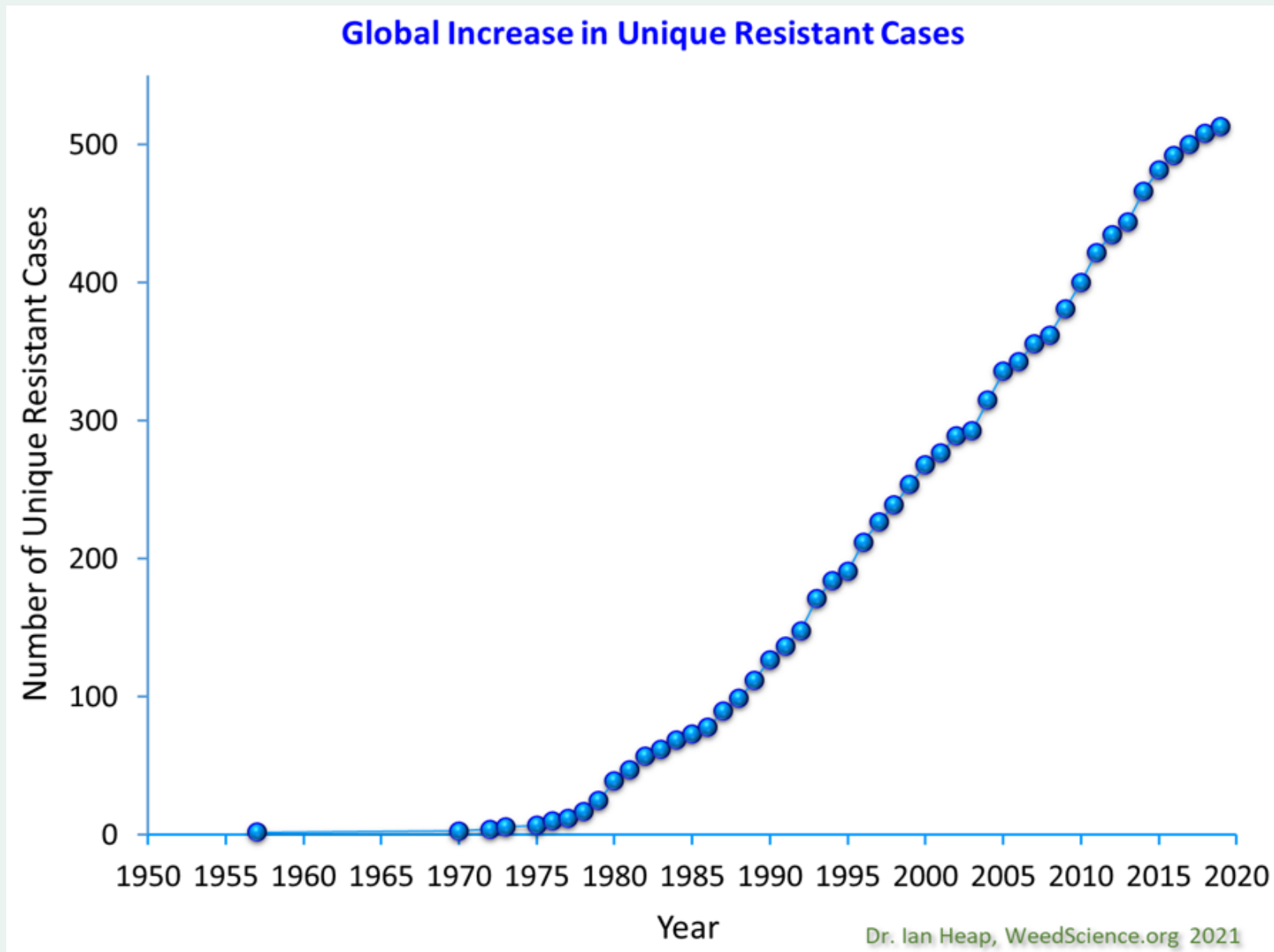
April 15, 2024



weed-management

For as long as growers have bottled weeds in their fields, weed

The Cost



Technology In Broad Acre Spraying

- 1800's – Hand & Horse drawn sprayers
- 1940's – Broad-acre boom sprayers
- 1950's-1960's: Larger and more efficient
- 1970's: Pressure and Boom improvements
- 1980's – Electronic rate controllers



Technology In Broad Acre Spraying

Before Rate Controllers

Change in Speed + Constant Pressure = Change in Rate

With Rate Controllers

Change in Speed + Constant Rate = Change in Pressure

- 1980's – Electronic rate controllers



Sprayer Technology



Technology In Broad Acre Spraying

- 1800's – Hand & Horse drawn sprayers
- 1940's – Broad-acre boom sprayers
- 1940's – Broad-acre boom sprayers
- 1950's-1960's: Larger and more efficient
- 1980's – Electronic rate controllers
- 1990's – GPS Guidance, Pulse Width Modulation[PWM] nozzle technology created
- 2000's – Auto-steer, and PWM nozzle technology expands
- 2010's –Adoption of PWM, larger sprayers with tanks 1000+ gal and booms up to 132'
- 2020's – AI-based targeted spraying

PWM Technology

- Independent Control of Rate and Pressure
- Control of individual nozzles



SymphonyNozzle



PWM provides the foundation for Vision Technology

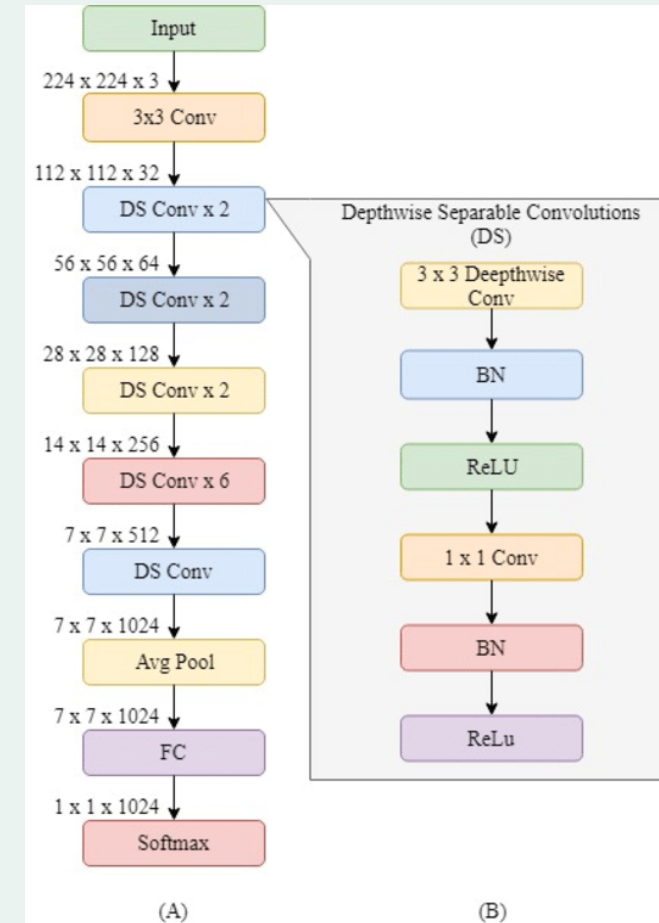
SymphonyVision



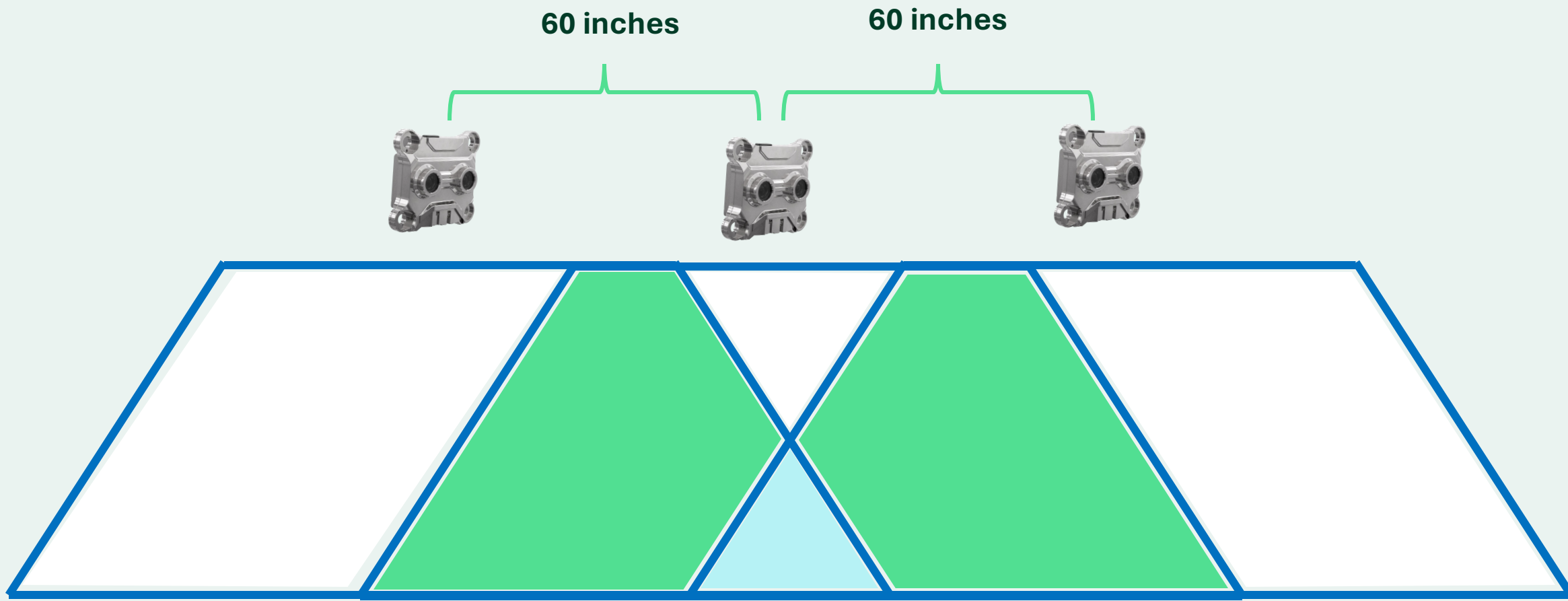
Model Overview

- 4 Channel Input - RGB + NIR
- Model Parameters: 9,764,484
- Model Layers: 358
- Runtime: ~38 ms

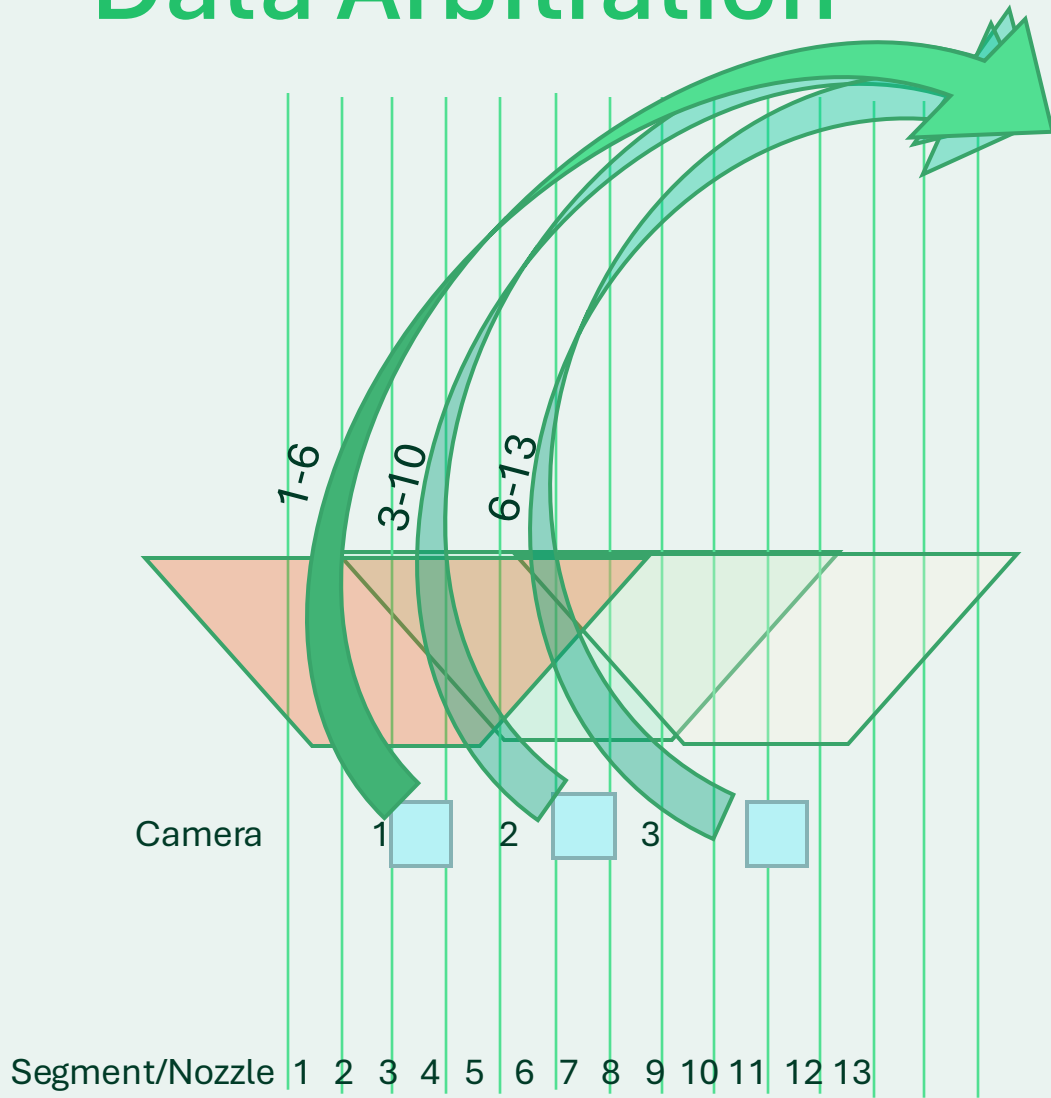
We manage all data collection and model training in house.



Camera View



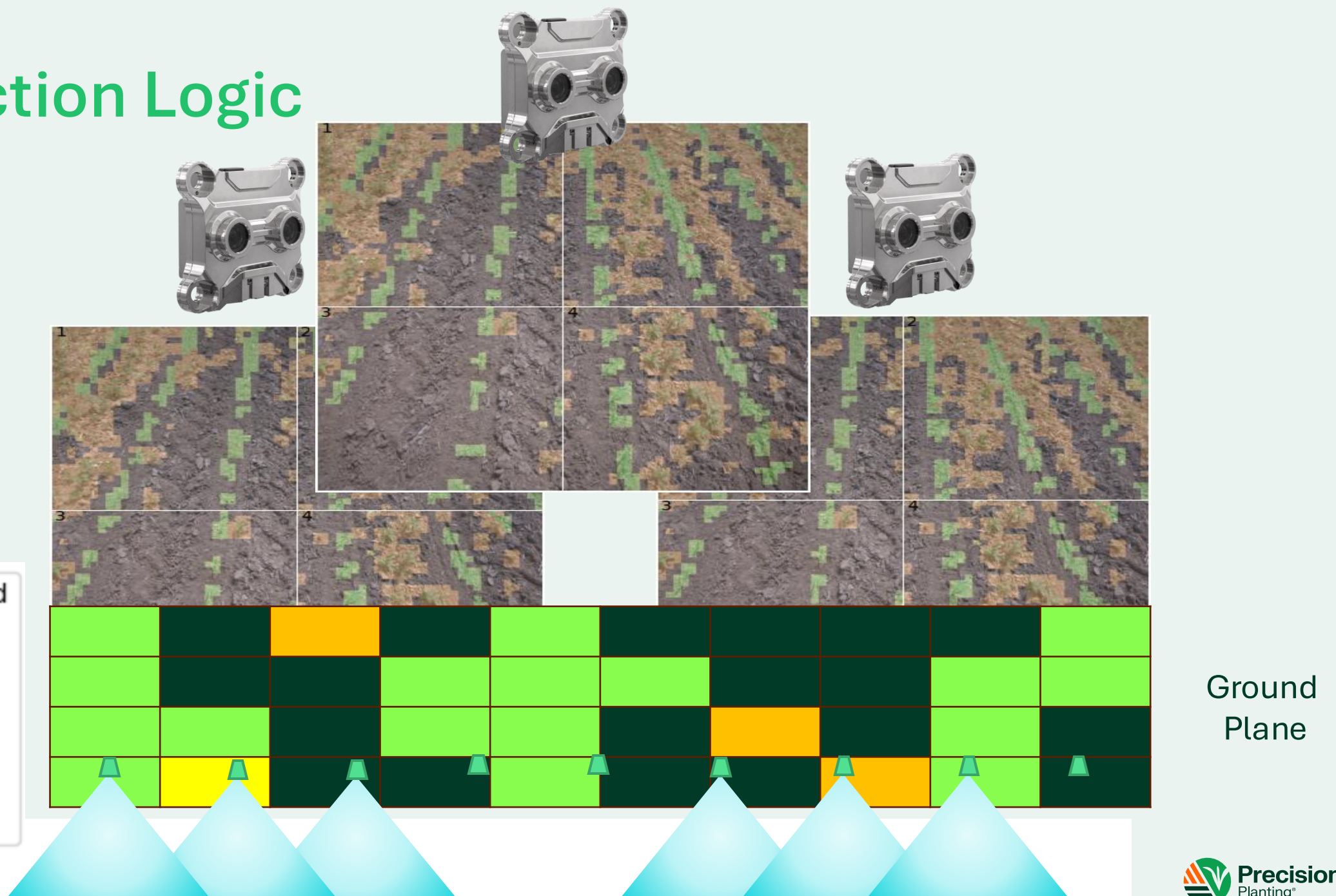
Data Arbitration



Symphony Nozzle



Detection Logic



Computer Vision – Feature Identification

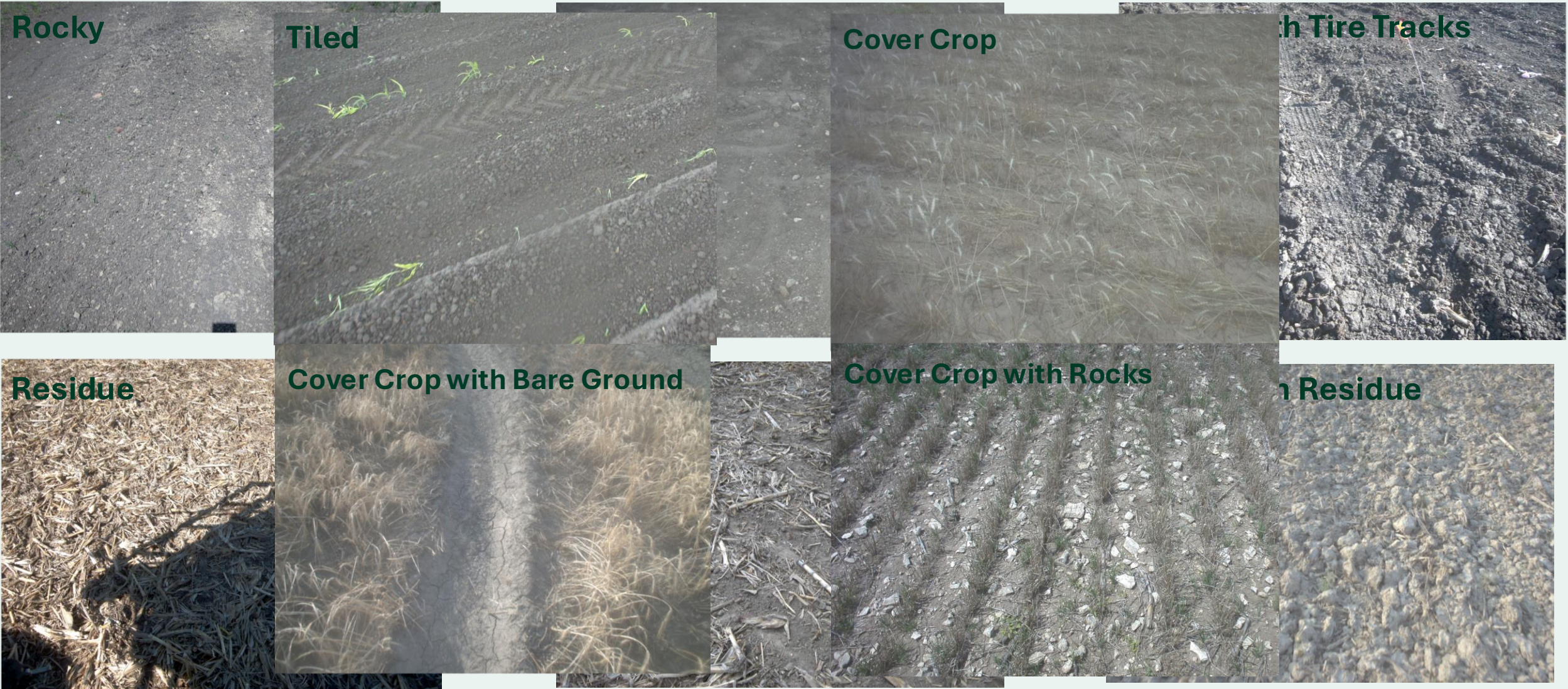


Green on Brown



Green on Green

Challenge: Backgrounds

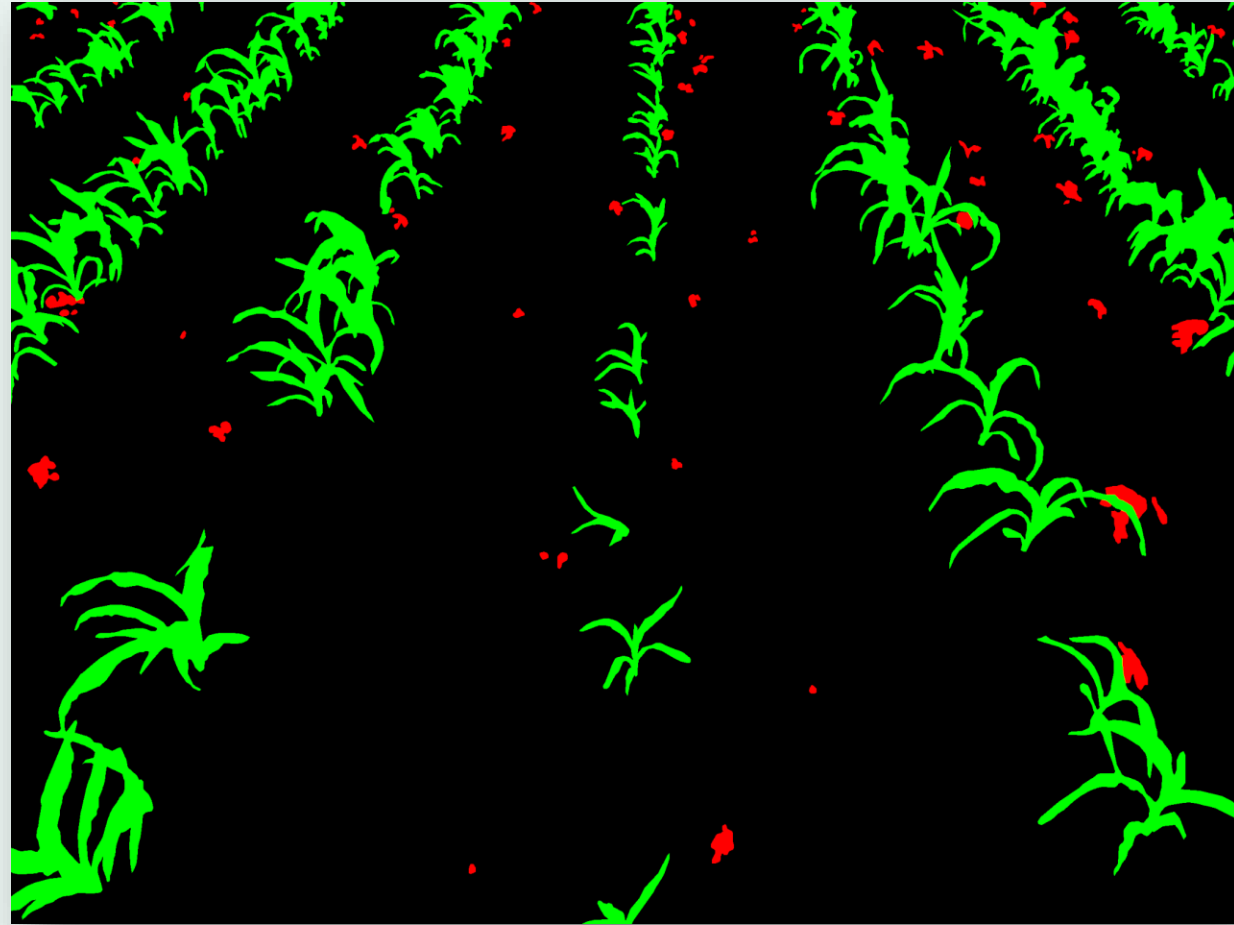


Challenge: Green on Green

Requires true leaf shape understanding



Challenge: Green on Green



Challenge: Green on Green

Requires true leaf shape understanding



Challenge: Time of Day



Challenge: Shadows



Challenge: False Positives From Run Over Crop



Challenge: False Positives from Shriveled Crop Leaves



RGB in Soybeans Prediction Side by Side



Weed Detection In-field Validation

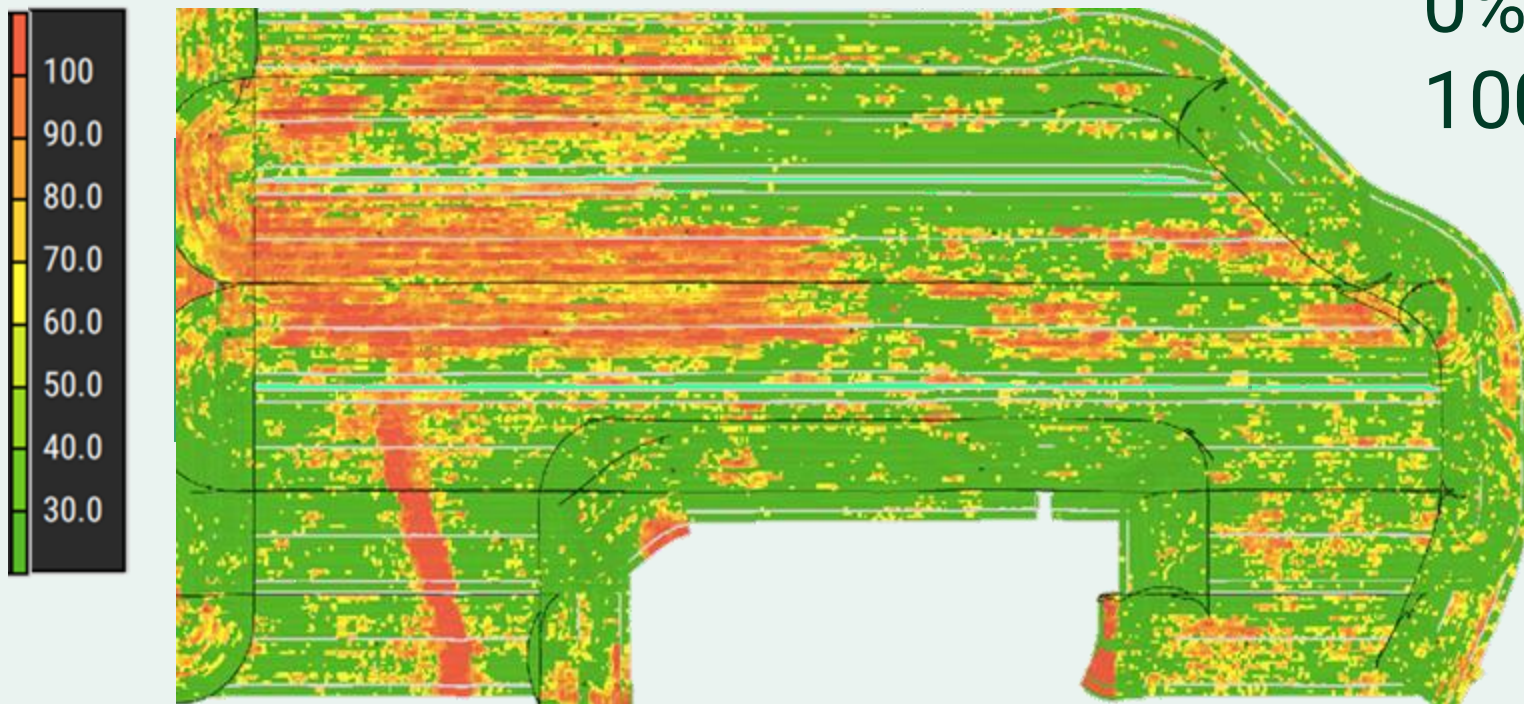


Shop 50 Weed Map – Weed Pressure

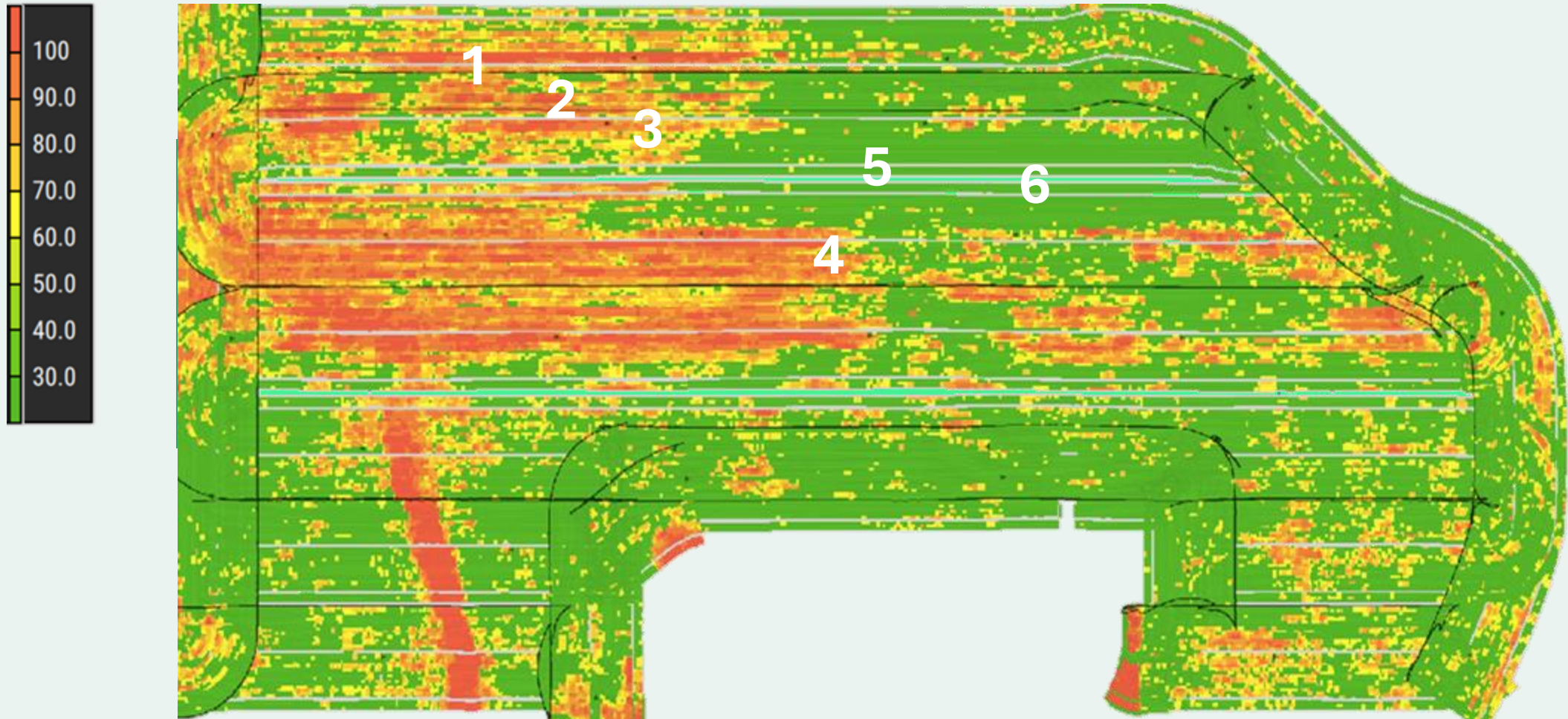
Weed Pressure

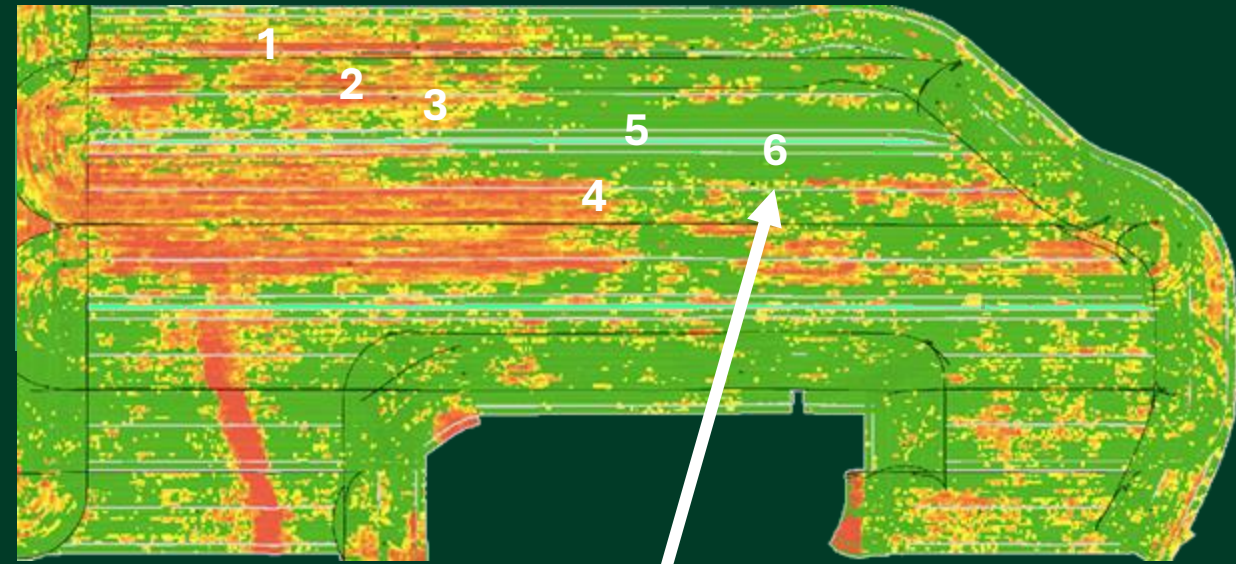
0% = no weed pressure

100% = High weed pressure

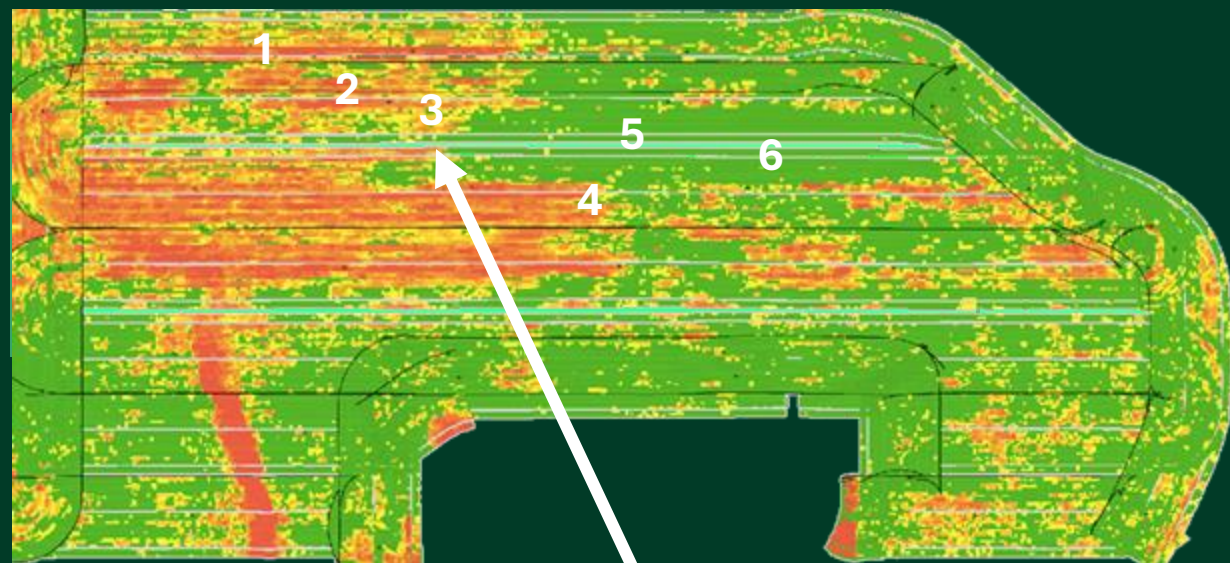


Shop 50 Weed Map – Spot Checks

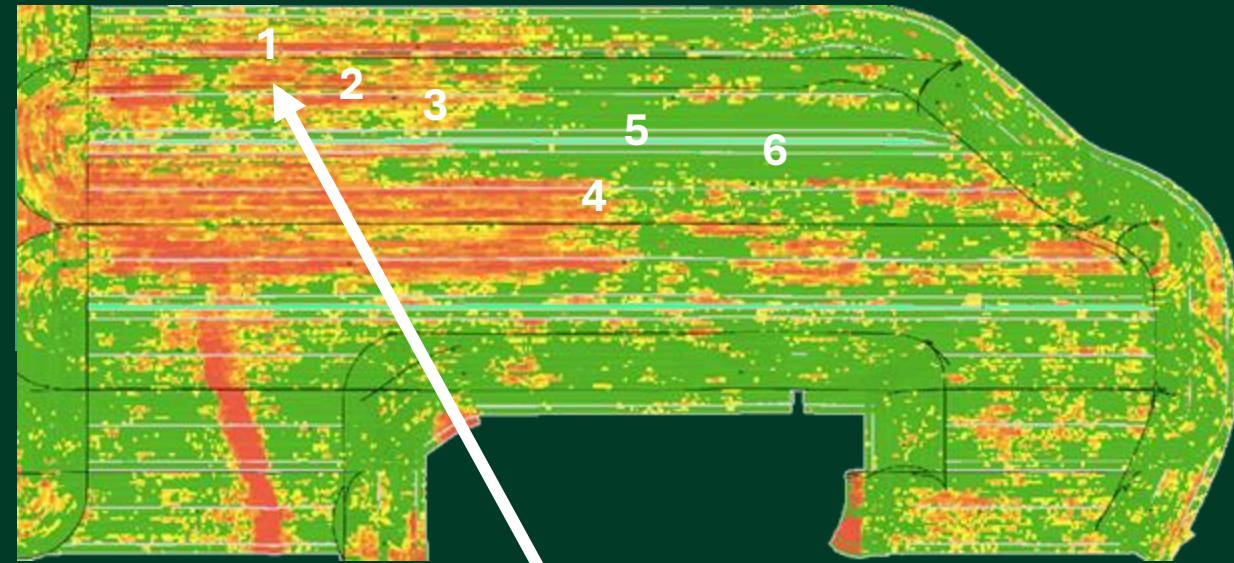




Spot check #6
No Weed
Pressure



Spot check #3
MEDIUM Weed
Pressure



Spot check #1
HIGH Weed
Pressure

Weed Density

- Percent of the ground covered in weeds.
- Green on Brown & Green on Green



Weed Type Classifications

Broadleaf



Grass



BL & Grass

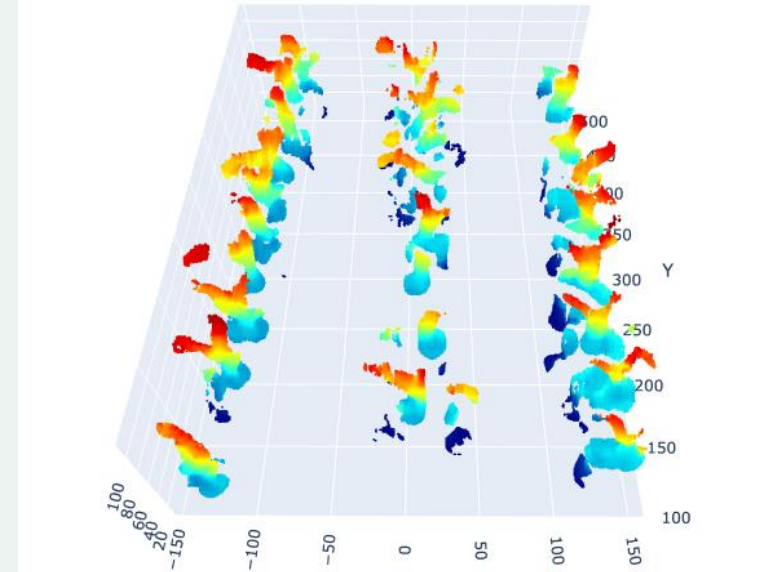


Emerging



Stereo Cameras

- Stereo camera can be used to get height information
- Utilize stereo vision to integrate height into understanding how severe the weeds are
- Map Crop Height



Weed Severity

- A value unique to Precision Planting that combines both weed density and weed height into a single value
- Farmer sees value as either: None, Low, Medium, or High



Current Compatible Crops

Corn



Soybeans



Cotton



Sorghum



Sugar Beets



Canola



Wheat*



Challenge: Grass in Wheat

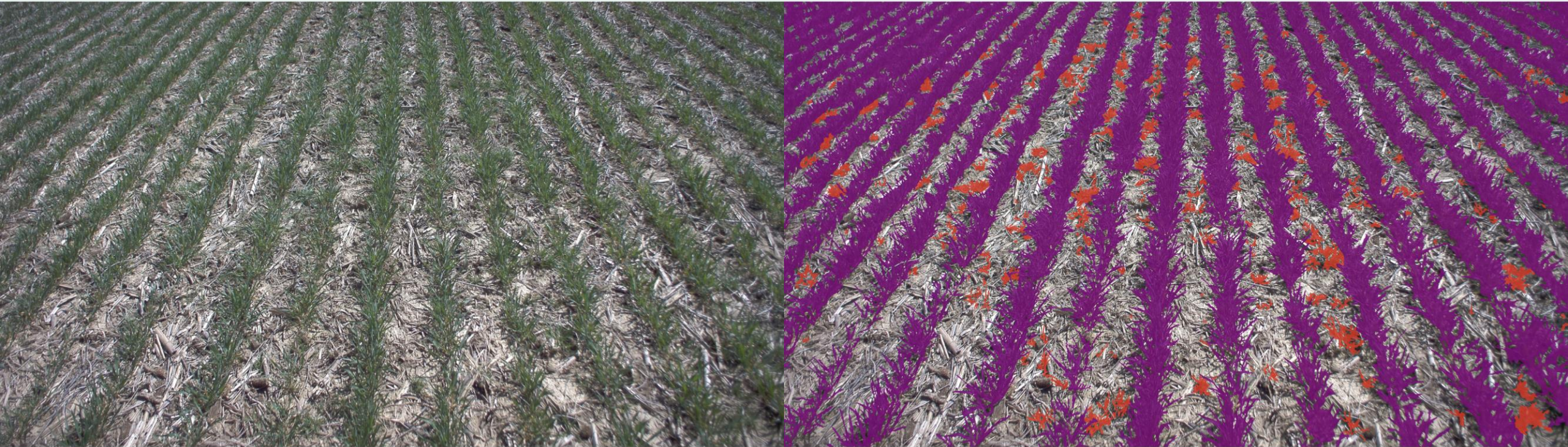


Wheat



Foxtail - grass

Challenge: Grass in Wheat



Challenge: Differing Growth Stages



Chemical Labels

- 2003: The United Nations formally adopted the Globally Harmonized System of Classification and Labeling of Chemicals [GHS]
- 2015 – 2016 Full compliance of manufactures and distributors in the United States
- Labels are legally binding and include information such as - safety/handling, identifiers, EPA info, ingredients, toxicity levels, and directions for use.

Adjuvant	• Ammonium sulfate (AMS) can be used at 1.5 lb/A to 3 lb/A. Rates are dependent on tankmix partners, environmental conditions, temperatures and potential for leaf burn. • AMS has shown to improve weed control of difficult-to-control weeds, like velvetleaf and lambsquarters, under difficult environmental conditions (low relative humidity) or hard water. • Anti-foam agent is recommended.
Surfactants/Oils	• The use of additional surfactants or crop oils may increase the risk of crop response. Please refer to surfactant label for more detailed directions.
Spray Volume	• 15 GPA minimum. • If dense canopy, large weeds or unfavorable growing conditions are present, increase water volume to a minimum of 20 GPA.
Nozzle Spray Quality	• Medium to Coarse nozzles. • LIBERTY 280 SL HERBICIDE is a contact herbicide and requires proper nozzles with uniform thorough spray coverage to achieve optimum weed control. • See nozzle section for more detailed information.
Rainfast	• 4 hours.
Maximum Number of Applications per Year	• Refer to Table 5

APPLICATION DIRECTIONS FOR USE ON LIBERTYLINK SOYBEANS

Apply LIBERTY 280 SL HERBICIDE only to soybean designated as LibertyLink. Uniform, thorough spray coverage is necessary to achieve optimum weed control.

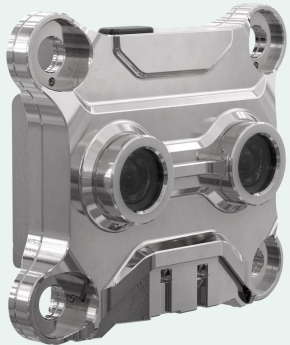
APPLICATION DIRECTIONS

Application Timing	• Apply to small and actively growing weeds, targeting less than 3 inch weeds in height. For additional information on weed heights refer to the Weed Control for Row Crops section. • For Best results, warm temperatures, high humidity, and bright sunlight improve the performance of LIBERTY 280 SL HERBICIDE. • Weed control may be reduced if application is made when heavy dew, fog, and mist/rain are present; or when weeds are under stress due to environmental conditions such as drought, cool temperatures, or extended periods of cloudiness. • To avoid the possibility of reduced lambsquarters, Palmer amaranth and velvetleaf control, applications should be made between dawn and 2 hours before sunset.
Application window	• Emergence up to bloom or R1 growth stage.
Application Use Rate	• Apply 32 - 43 fl oz/A depending on weed species, size and density per weed chart. • If required a second application of 32 - 43 fl oz/A can be applied up to a yearly maximum of 87 fl oz/A. • The second application should be made minimum 5 days after the first application.
Maximum Per Year	• 87 fl oz /A.





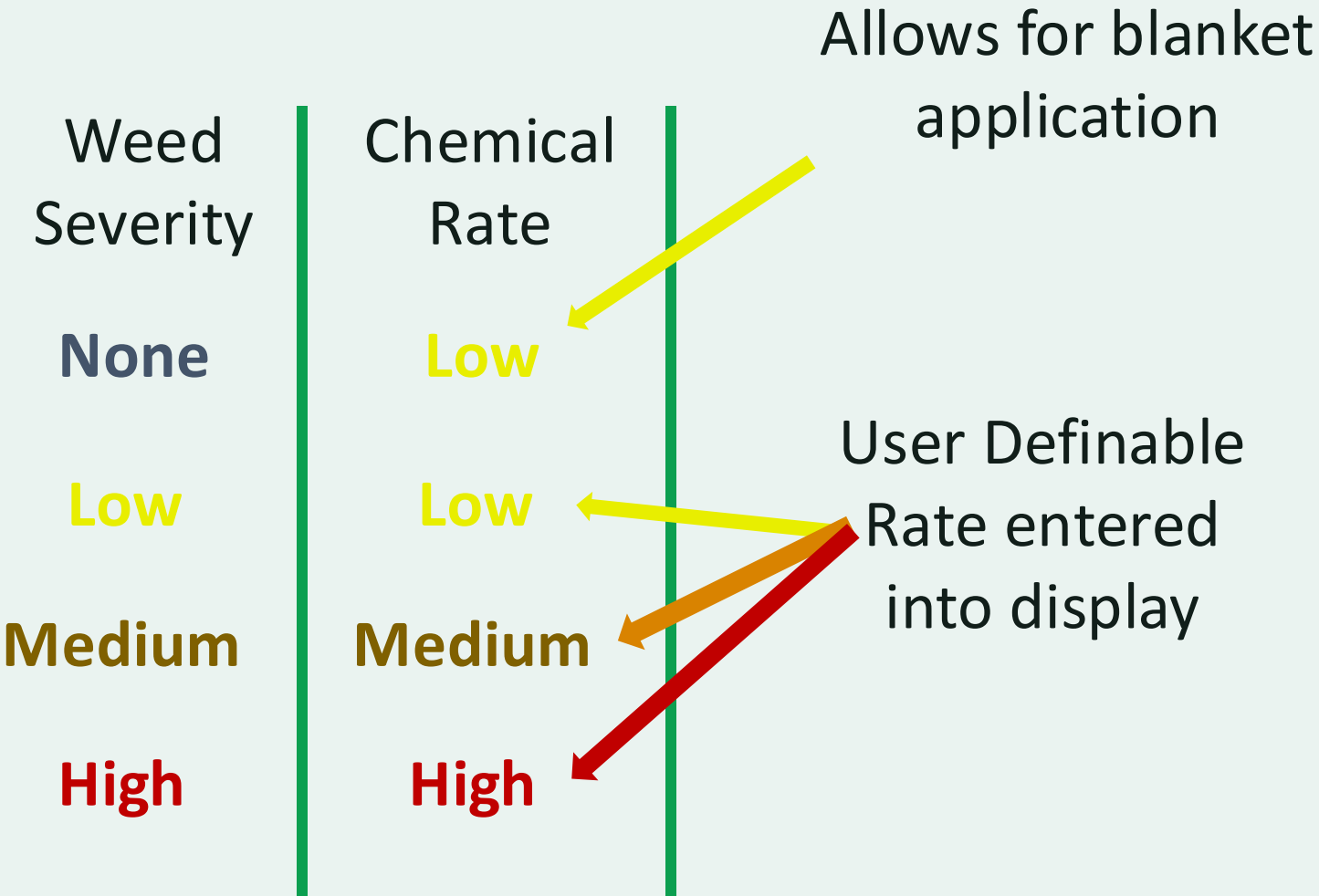
SymphonyVision | Rate



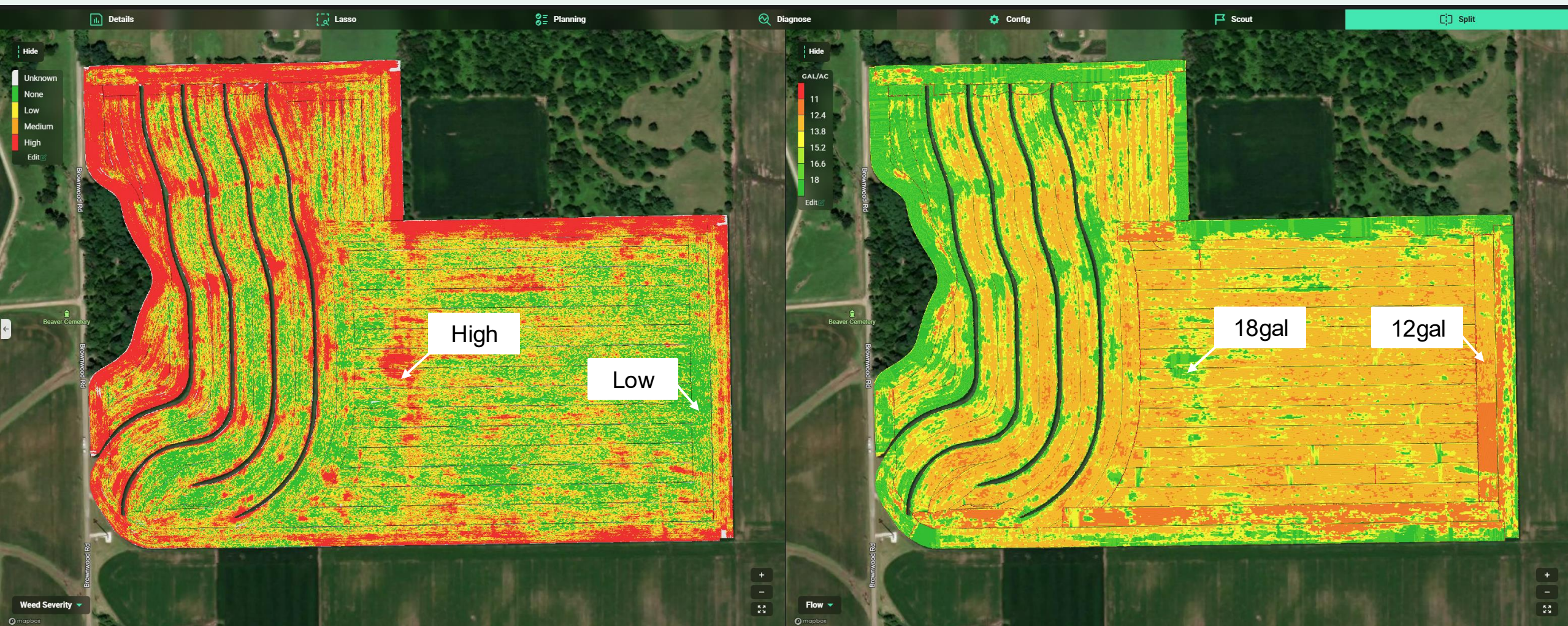
- Automatically adjust the applied rate based on Weed Severity
- Continuous base rate for full field coverage
- ~10' camera spacing



SymphonyVision | Rate



SymphonyVision | Rate



Weed Severity Map

Flow Map

SymphonyVision | Spot

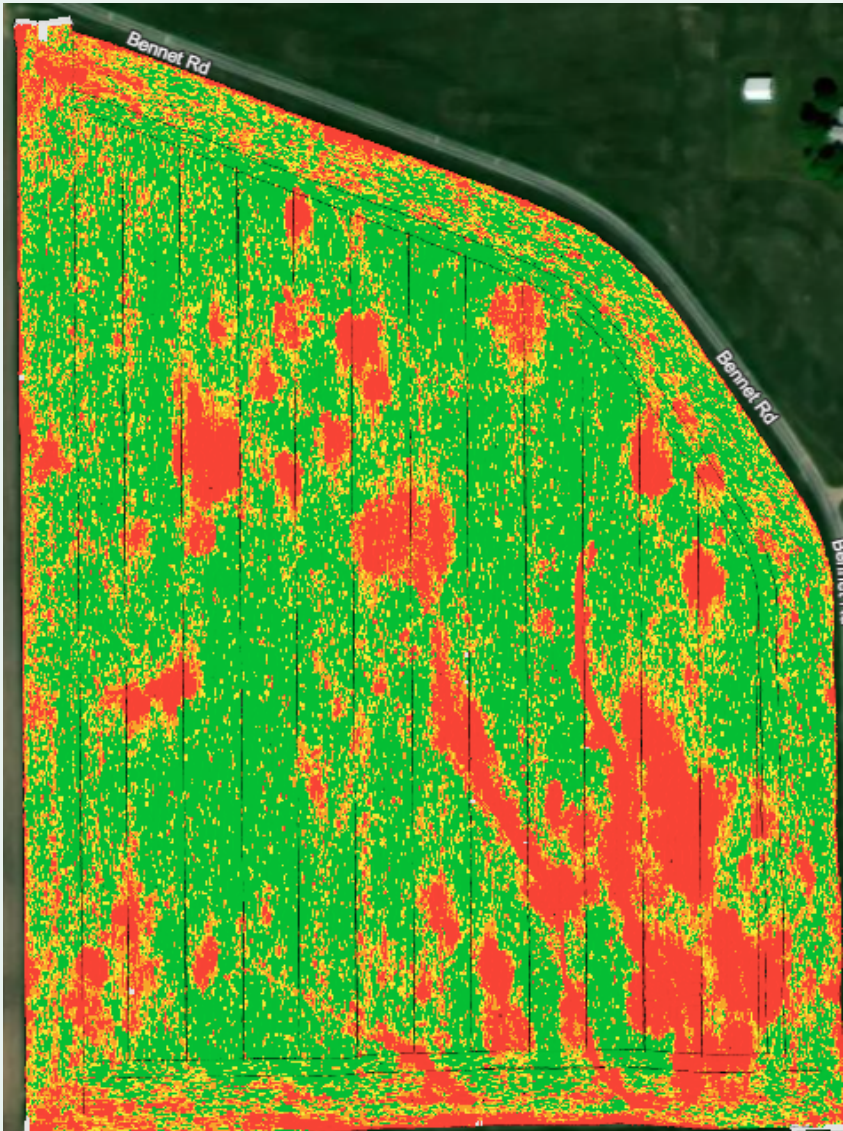


- Automated Variable Rate **Spot** Spray
- Turns Off Nozzle When Weeds are Not Present
- Improved Accuracy (Especially In-Crop)
- (5' Camera Spacing)





Post Pass 1



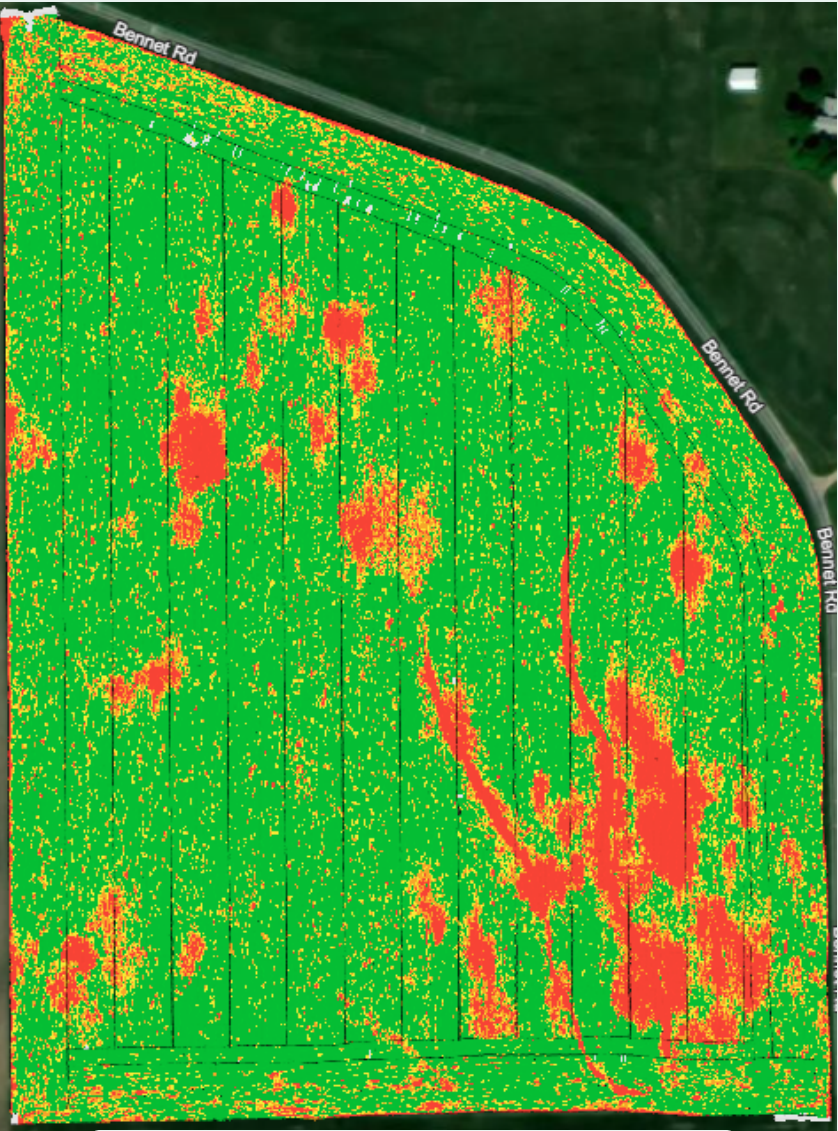
Weed Severity Map

Weed Severity			
None	Low	Medium	High
52.7%	11.7%	16.3%	19.3%

Field	43.5 acre
Target Rate	20 gal/ac
Cost to Blanket Spray	\$22.43/acre
Gallon Usage to Blanket Spray	870 gal

Percent of Target Rate Sprayed for Weed Severity			
None	Low	Medium	High
0%	100%	100%	100%
Actual Cost to Spray			\$15.39/acre
Actual Gallons Used			597 gal
Savings			31.4%
\$ Savings with SymphonyVision			\$7.04/acre

Post Pass 2



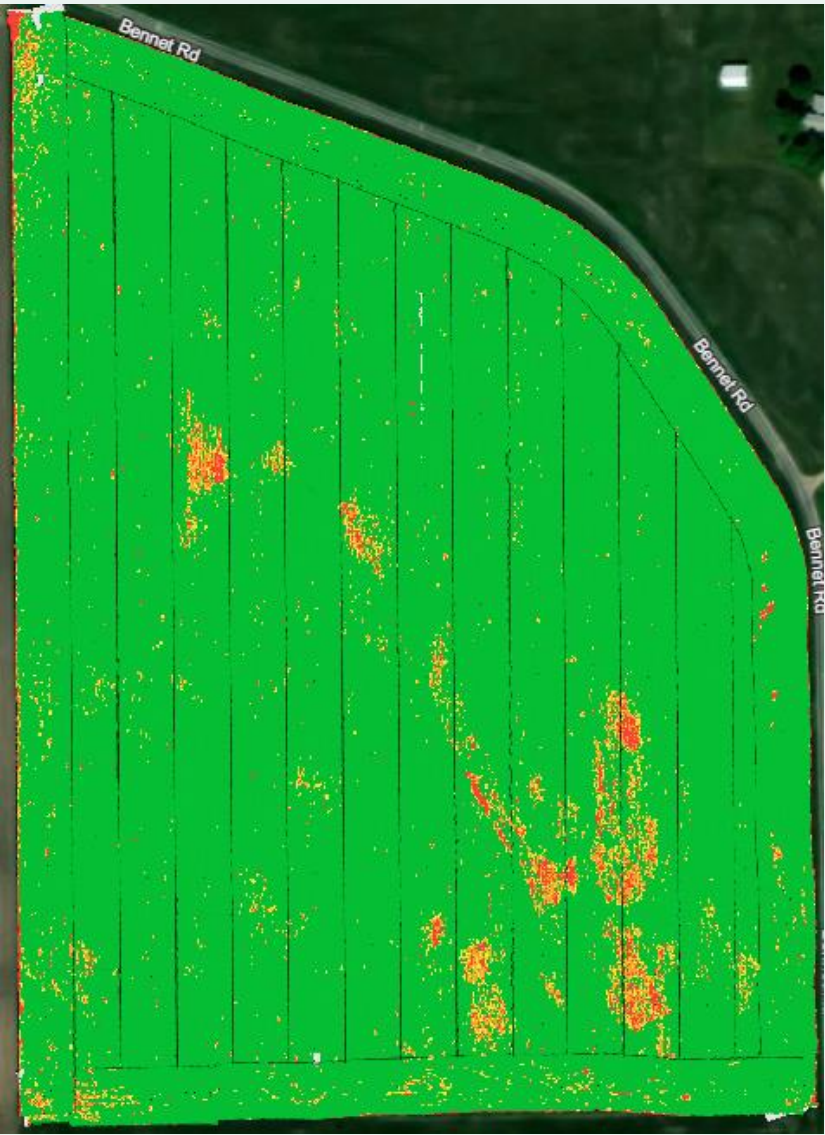
Weed Severity Map

Weed Severity			
None	Low	Medium	High
71.3%	9.7%	8.8%	10.2%

Field	43.5 acre
Target Rate	24 gal/ac
Cost to Blanket Spray	\$24.12/acre
Gallon Usage to Blanket Spray	1044 gal

Percent of Target Rate Sprayed for Weed Severity			
None	Low	Medium	High
0%	100%	100%	100%
Actual Cost to Spray			\$11.67/acre
Actual Gallons Used			505 gal
Savings			51.6%
Savings with SymphonyVision			\$12.45/acre

Post Pass 3 – (Unplanned)



Weed Severity Map

Weed Severity			
None	Low	Medium	High
94.5%	1.9%	2.2%	1.4%

Field	43.5 acre
Target Rate	25 gal/ac
Cost to Blanket Spray	\$12.60/acre
Gallon Usage to Blanket Spray	1088 gal

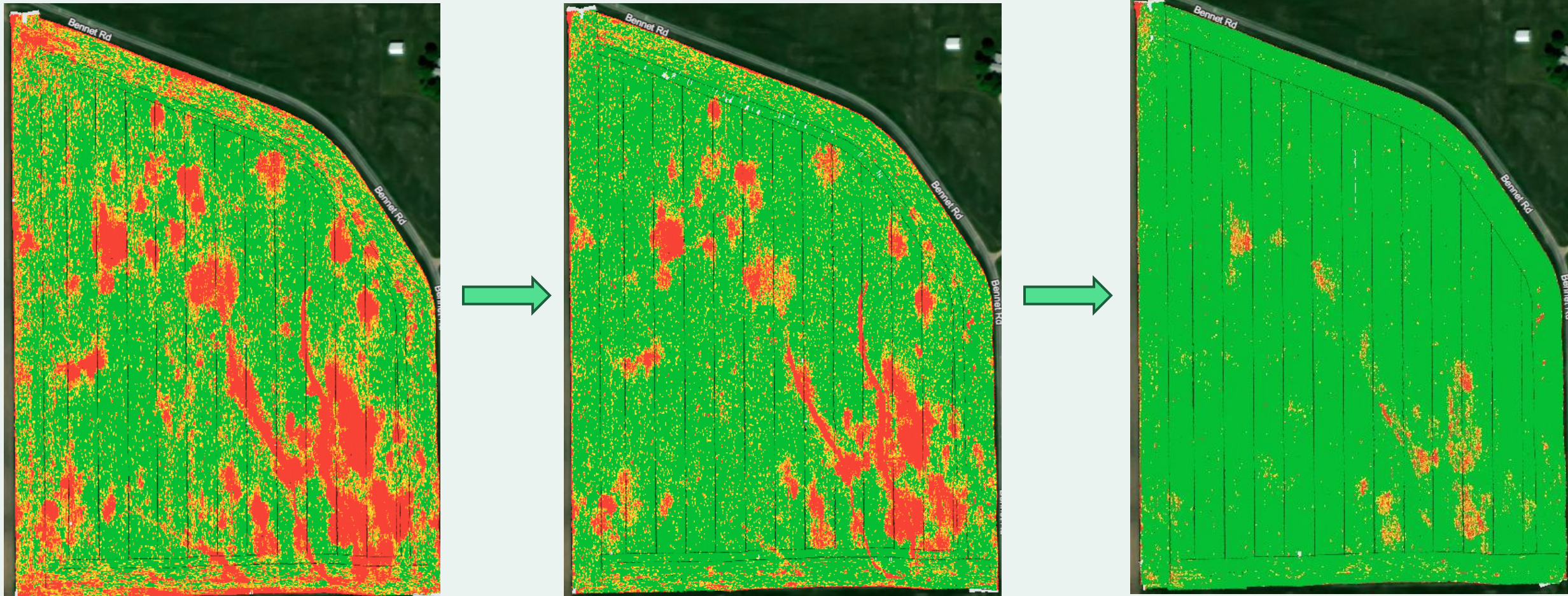
Percent of Target Rate Sprayed for Weed Severity			
None	Low	Medium	High
0%	100%	100%	100%
Actual Cost to Spray			\$1.45/acre
Actual Gallons Used			125 gal
Savings			88.5%
Savings with SymphonyVision			\$11.15/acre

Costs

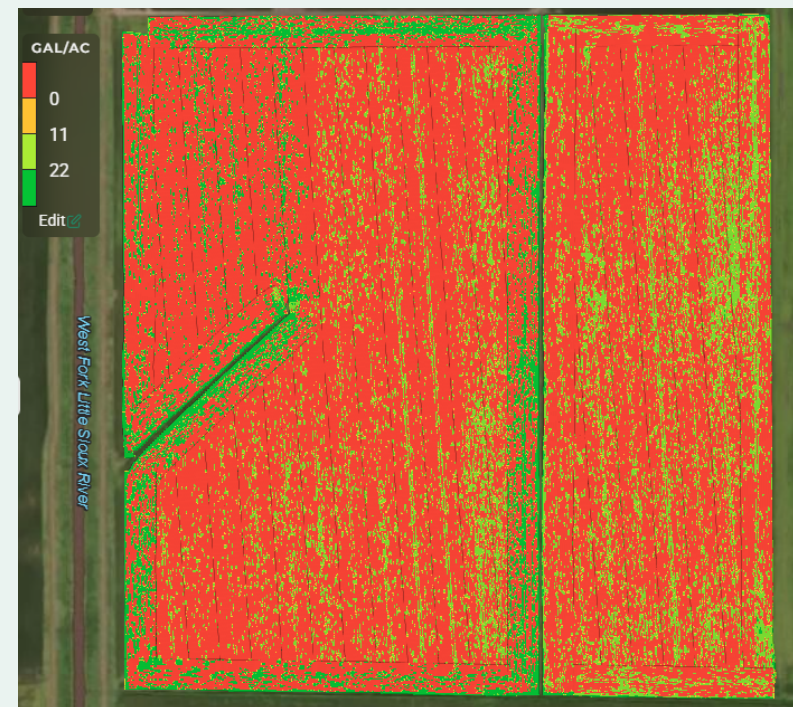
Original Planned Cost for TWO Post Passes	\$46.55 per acre
Actual Cost to Spray THREE Post Passes	\$28.51 per acre

Gallons over 3 passes

To Blanket Spray	3,002 gal
Actual	1,227 gal



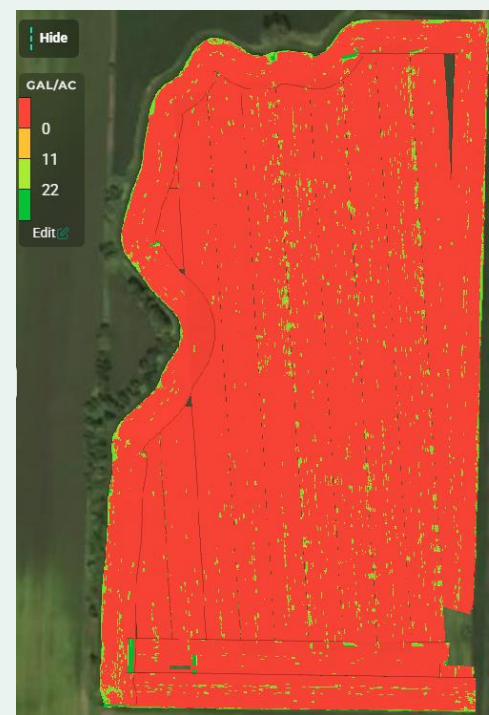
Weed severity maps across three passes in the field



Field 1

Gallons to Blanket
Spray – 2,733

Gallons Sprayed - 535



Field 3

Gallons to Blanket
Spray – 1,451

Gallons Sprayed - 60

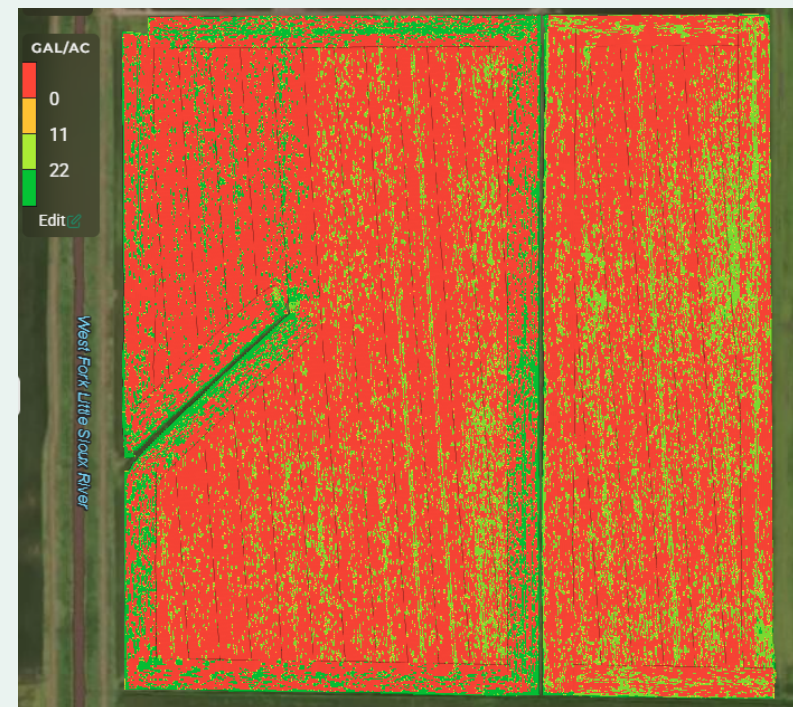


Field 2

Gallons to Blanket
Spray – 307

Gallons Sprayed - 50

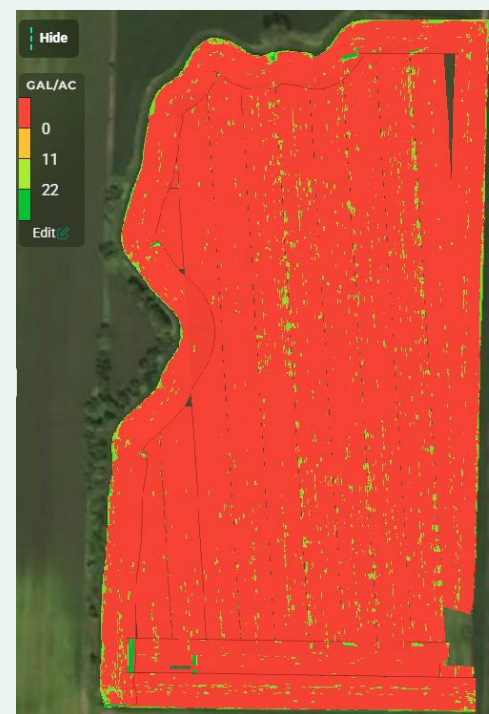
Weed Severity	Percent of 21.5 gal/ac Target Rate
None	0%
Low	100%
Medium	100%
High	110%



Field 1

Gallons to Blanket
Spray – 2,733

Gallons Sprayed - 535



Field 3

Gallons to Blanket
Spray – 1,451

Gallons Sprayed - 60



Field 2

Gallons to Blanket
Spray – 307

Gallons Sprayed - 50

Total Acres	208.9
Gallons to Blanket Spray	4,491 gal
Actual Gallons Used	645 gal
Percent Saved	85.6%

SymphonyVision | Duo





**Residual
+
Contact**

Residual



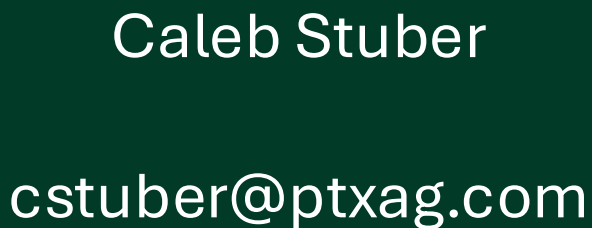
Where is the ROI with SymphonyVision?

- Chemical Reduction
- Refilling less often
 - Time Savings
 - Less engine hours
 - Less fuel
- Yield Bump from crop not metabolizing chemicals
- Ability to variable rate based on weed severity
 - Spray higher rates on larger plants and save in areas with no weeds
- Allowing growers to rethink their weed control program
- Maps
 - Scouting Fields
 - Reporting
 - Future Decision Making

What Does The Future Look Like?

- Detection of Crop Diseases?
- Nutrient Deficiencies in Crops?
- Types/Amounts of Insects?
- Identifying Weed Types?
- Steer Sprayers Using Cameras to Reduce Running Over Crop

We want to give farmers the tools they need to better manage their fields while reducing chemical amounts and be the best stewards that they can.

[illegible]